



Letter of Collaboration

26 August 2026

To the National Science Foundation,

I am writing on behalf of Imatest LLC to express our interest in collaborating with the FreeMoCap Foundation if it is selected for funding through the NSF Pathways to Enable Secure Open-Source Ecosystems (PESOSE) program.

Imatest develops software and test methods for measuring and characterizing the performance of cameras and imaging systems. Our work focuses on enabling designers and users of imaging systems to understand how camera hardware, image processing, and other parts of the imaging pipeline affect the quality and information content of captured images. In addition to my work at Imatest, I am involved in the ongoing ISO/TC 42/WG 18 work related to image-information content (ISO/WD 23654).

There is a natural overlap between this work and some of the challenges FreeMoCap faces as an open source markerless motion capture system. FreeMoCap is intentionally hardware-agnostic and can be used with relatively inexpensive consumer cameras. This raises important questions about what makes a camera suitable for computer-vision-based motion measurement and how those characteristics should be measured and reported. Users of FreeMoCap need to be able to distinguish limitations introduced by the imaging system from limitations introduced later in the computer-vision and processing pipeline.

If funded, we would be interested in continuing discussions with the FreeMoCap team around these questions and exploring areas where Imatest's experience with image and video quality measurement may be useful. This could include sharing our perspective on relevant spatial and temporal imaging characteristics, camera characterization methods, and applicable standards, as well as exploring how camera characterization might inform areas such as hardware recommendations or benchmarking practices. The nature and scope of the collaboration would depend on the needs of the project and areas of mutual interest.

My involvement in ongoing standards work related to image-information content may also provide a useful perspective for these discussions. The questions that FreeMoCap is asking about camera suitability and measurement quality have clear connections to broader questions about how the information produced by imaging systems can be characterized and evaluated. FreeMoCap may also provide an interesting real-world context for considering some of the practical challenges that arise when these concepts are applied to motion capture.

We see value in bringing an imaging-system perspective to FreeMoCap's broader work. Considering the role of camera hardware alongside the computer-vision and processing pipeline may help users and contributors better understand the factors that affect motion-capture measurements and make more informed decisions about cameras, acquisition setups, and downstream analysis.

Sincerely,

Sarah Kerr
Imaging Scientist
Imatest LLC

Relationship to proposing team: External technical collaborator